

Well-pointed objects, capitalization, and why Yoneda isn't enough

Freyd & Scedrov, §1.522–§1.547

From support (§1.522) and well-pointedness (§1.523) up through the §1.54 capitalization, with the comparison to the Yoneda embedding: when point-wise reasoning is valid, why “just embed into Set” is a theorem not a freebie, and why slicing (a pullback) stays exact where Yoneda (a hom) does not. Q&A distilled from a discussion, with SET/I fiber-product pictures.

§1. Projectives, and a group as a one-object category (§1.524)

A warm-up: projectivity (§1.524) — where SET is special and structure breaks it — and the “a group is a one-object category” view. Both recur below.

Definition 1.1 (Projective (§1.524)).

An object P is **projective** — three equivalent forms:

1. **Lifting.** Every cover $e : A \twoheadrightarrow B$ and every map $f : P \rightarrow B$ admit a **lift** $g : P \rightarrow A$ with $g e = f$ (diagram order: first g , then e).
2. **$\text{Hom}(P, -)$ preserves covers.** It carries every cover (surjection) to a surjection — so P survives quotients without losing information.
3. **Covers onto P split.** Every cover $e : A \twoheadrightarrow P$ has a section $s : P \rightarrow A$ ($s e = \text{id}_P$).

(Why “projective”: P is projective $\iff P$ is a **direct summand of a free object** ($F \cong P \oplus Q$), i.e. the **image of a projection** — an idempotent $e : F \rightarrow F$, $e e = e$. Take a free $q : F \twoheadrightarrow P$; lifting id_P along q gives a section, so P is a retract of F . Named as the dual of **injective** (Cartan–Eilenberg).)

In SET **every** object is projective: a cover is a surjection, so pick one preimage in each fiber (choice). That picked function is **automatically** a morphism — in SET a morphism **is** just a function. It stops being free the moment morphisms carry structure.

Counterexample — $\mathbb{Z}/2$ in $\mathbf{Ab}$. Covers in $\mathbf{Ab}$ are the surjective homomorphisms. Take the cover $q : \mathbb{Z}/4 \twoheadrightarrow \mathbb{Z}/2$ ($1 \mapsto 1$) and $f = \text{id} : \mathbb{Z}/2 \rightarrow \mathbb{Z}/2$. A lift would be a **homomorphism** $g : \mathbb{Z}/2 \rightarrow \mathbb{Z}/4$ with $g q = \text{id}$:

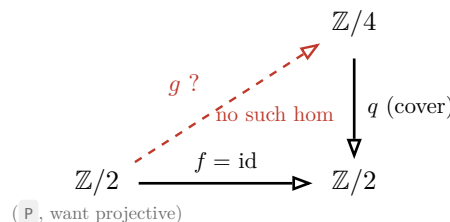


Figure 1: Lifting test for $P = \mathbb{Z}/2$. The cover $q : \mathbb{Z}/4 \twoheadrightarrow \mathbb{Z}/2$ and $f = \text{id}$ admit no homomorphism g with $g q = \text{id}$.

A homomorphism $\mathbb{Z}/2 \rightarrow \mathbb{Z}/4$ must send the generator to an element of order dividing 2, i.e. to 0 or 2; either way q sends it to $0 \neq 1$. So $g q$ kills the generator and cannot be $\text{id} : \mathbb{Z}/2$ is **not projective**.

★ **Punchline**

The SET “pick a preimage” still works **set-theoretically** ($1 \mapsto 1, 0 \mapsto 0$), but that function is **not a homomorphism** ($1+1=0$ in $\mathbb{Z}/2$, yet $1+1=2 \neq 0$ in $\mathbb{Z}/4$). Projectivity needs the lift to be a **morphism** of the category; choice only hands you a bare function — that gap is “not projective.” The projectives of **Ab** are exactly the **free** abelian groups, and $\mathbb{Z}/2$ is not free.

§1.1. A group is a one-object category (a groupoid)

$\mathbb{Z}/2, \mathbb{Z}/3$ are also interesting **as whole categories** — a different role from being objects of **Ab**. Any group **G** is a category with **one object** $*$; **morphisms** $* \rightarrow *$ = the group elements; **composition** = the group operation; **identity** = the unit. Every morphism is invertible, so it is a **groupoid**, and its composition table **is** the group’s Cayley table. For $\mathbb{Z}/2$ that is: one object $*$, the identity, and a single non-identity iso r with $r \cdot r = \text{id}$ (so $r^{-1} = r$); for $\mathbb{Z}/3$: the identity and a mutually-inverse pair r, r^2 ($r^{-1} = r^2$).

The relations live in the geometry once you **unroll** the self-loops into a walk: every node is the **same** object $*$, tagged by which power of r has acted, and a walk that returns to the start **is** the identity. So $r \cdot r = \text{id}$ is a closed 2-cycle and $r \cdot r \cdot r = \text{id}$ a closed triangle (the group’s Cayley graph).

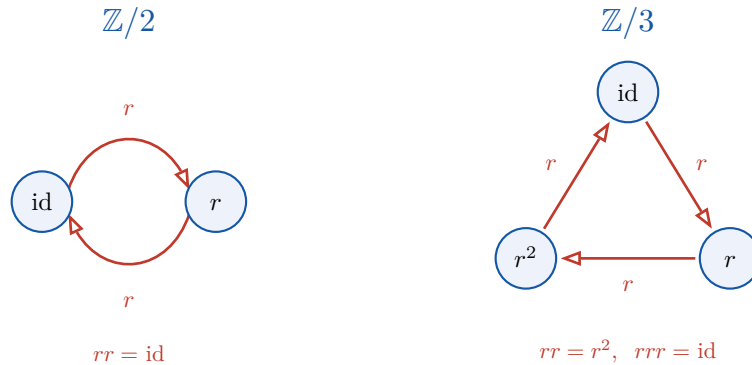


Figure 2: The relation read off the diagram: a closed walk of r ’s = **id**. $\mathbb{Z}/2$ closes in two steps ($r \cdot r = \text{id}$); $\mathbb{Z}/3$ in three ($r \cdot r = r^2, r \cdot r \cdot r = \text{id}$). Each node is the **same** object $*$, tagged by which power of r has acted; collapsing all nodes back to one $*$ gives the one-object groupoid.

§2. Support: the image of $A \rightarrow 1$ (§1.522)**Definition 2.1** (Support, well-supported).

In a regular category the **support** $\text{Spt}(A)$ is the **image of the unique map** $A \rightarrow 1$ — always a **subobject of 1**. A is **well-supported** if $\text{Spt}(A) = 1$; equivalently (pre-regular) $A \rightarrow 1$ is a **cover**. So A is *not* well-supported $\iff A \rightarrow 1$ factors through a *proper* subobject $U \subsetneq 1$ (§1.534: $A \rightarrow U \rightarrow 1$).

Support lives among the **subobjects of 1** (the “truth values” / support lattice). How many there are controls how interesting the notion is.

Category	Subobjects of 1	$\mathbf{Spt}(A)$	Not well-supported $\iff$	Example
SET	just $\emptyset, \{*\}$	1 if $A \neq \emptyset$, else $\emptyset$	$A = \emptyset$	$\emptyset$ — unique in Set
Set/I	subsets of I	image $p(A) \subseteq I$ of $p: A \rightarrow I$	p misses some $i \in I$ (empty fiber)	$I = \{0, 1\}$, $\{*\} \mapsto 0: \{0\} \subsetneq I$
Sh(X)	open sets of X	open set where sheaf $\neq \emptyset$	sheaf on proper open $U \subsetneq X$	extend-by- $\emptyset$ off U

Set picture ($1 = \{*\}$ has only two subobjects): $A \neq \emptyset$ gives $A \rightarrow 1$ surjective, image $= 1$, **well-supported**; $A = \emptyset$ gives image $\emptyset \subsetneq 1$, **not** well-supported. So $\emptyset$ — also the *initial* object — is the unique not-well-supported object of SET. (No zero object is involved: in Set initial $\emptyset \neq \{*\}$ terminal.)

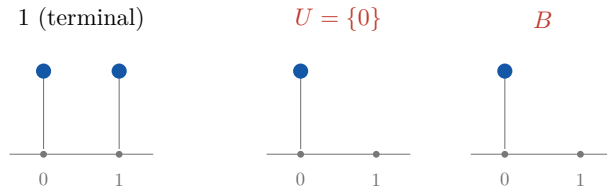


Figure 3: Objects of **Set/I** as dots fibered over the base $I = \{0, 1\}$. The terminal 1 occupies *both* fibers; the proper subobject $U = \{0\}$ and the not-well-supported B (red) occupy *only* fiber 0 . B 's image in I is $\{0\} \subsetneq I$.

The consequence (§1.533–§1.534) is that slicing over a **not**-well-supported B throws information away, so capitalization slices only over **well-supported** B (also the hypothesis of *capital*, §1.525 — see the next section). Here is the reason, once, in the scene above.

§2.1. Why slicing over a not-well-supported B collapses (§1.533–§1.534)

The product is a fiber product. $B \times U$ is the product *in the ambient category*; in **Set/I** that is the **fiber product over I** (a **pullback**), computed as the **ordinary product inside each fiber**:

$$(B \times_I U)_i = B_i \times U_i.$$

(In plain **Set** it would be the ordinary Cartesian product — but there 1 has only $\emptyset, \{*\}$, so the sole proper U is $\emptyset$ and §1.534 is degenerate; the slice is what gives 1 many subobjects, the subsets of I .)

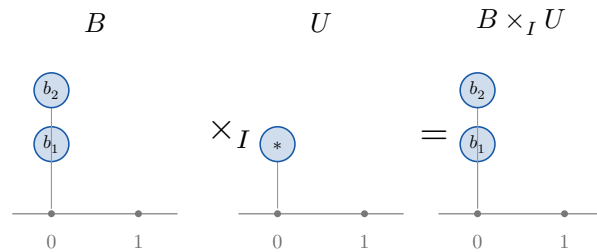


Figure 4: Fiber product $B \times_I U$ over $I = \{0, 1\}$. B lives only over 0 ($B_0 = \{b_1, b_2\}$, $B_1 = \emptyset$); the subterminal $U = \{0\} \rightarrow 1$ is a single point over 0 ($U_0 = \{*\}$, $U_1 = \emptyset$). Fiberwise $(B \times_I U)_i = B_i \times U_i$: over 0 , $\{b_1, b_2\} \times \{*\} = \{b_1, b_2\}$ (a fiber $\times$ a point is unchanged); over 1 , $\emptyset \times \emptyset = \emptyset$.

So $B \times_I U \cong B$.

Why the map is an isomorphism. U is a single point $\{*\}$ over its support and $\emptyset$ elsewhere, so $\times U$ restricts to U 's support: a fiber $\times \{*\}$ is unchanged, a fiber $\times \emptyset$ is emptied. B already lives over $\emptyset$ (not well-supported, $\text{Spt}(B) = \{\emptyset\} \subseteq U$), so nothing is dropped — which is §1.534:

§1.534 (verbatim). Suppose that B is not well-supported, that is, there exists $B \rightarrow U \rightarrow 1$ where U is a proper subobject of 1 . Then $B \times U \rightarrow B \times 1$ is an isomorphism and hence $\Delta(U) \rightarrow \Delta(1)$ is an isomorphism. That is, Δ is not faithful.

Why this collapses Δ . $\Delta(X) = (X \times B \rightarrow B)$ shows X only through B 's support (fiber $\emptyset$); it is blind to fiber 1 . So two sections of a Y that agree over $\emptyset$ but differ over 1 become indistinguishable:

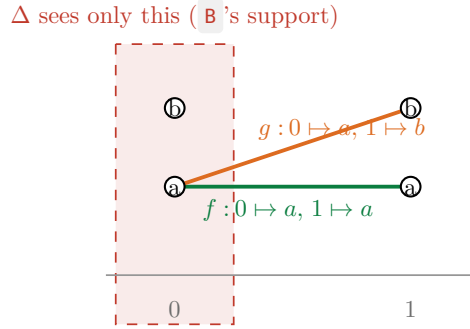


Figure 5: Y has fibers $\{a, b\}$ over each point. Sections f (green) and g (orange) agree over $\emptyset$ (both pick a) and differ over 1 . Slicing over B looks through the shaded column (B 's support) only, so Δ records just $f(\emptyset)=g(\emptyset)=a: \Delta f = \Delta g$ though $f \neq g$. So Δ is **not faithful** exactly when B is not well-supported.

★ Punchline

Slicing over B = “looking at the whole category through B .” If B exists only over part of the world, everything outside B 's support is invisible — distinct subobjects (U vs 1) and distinct maps (f vs g) that differ only out there get identified. That is why capitalization slices only over **well-supported** B ($B \rightarrow 1$ a cover, B touches *every* point): only then does Δ 's “look through B ” miss nothing, and the embedding loses no morphism.

§3. Well-pointed (§1.523)

Definition 3.1 (Well-pointed).

A is **well-pointed** if the global points $1 \rightarrow A$ **jointly cover** A : the only subobject containing every point $1 \rightarrow A$ is A itself. Elementary form: a mono $A' \rightarrow A$ through which every point $1 \rightarrow A$ factors must be an isomorphism.

Set intuition: “ A is covered by its elements” — nothing in A is invisible to the points. (Distinct from but cousin to “ 1 is a separator” = $\text{Hom}(1, -)$ faithful.)

§4. Which categories are well-pointed

Category	WP?	A point $1 \rightarrow A$ is...	Why / witness
Set, FinSet	✓ all	an element	a subset with every element is A . The canonical well-pointed topos.
Grp	✗	hom from trivial group $\Rightarrow$ identity only	zero object ($1 \cong 0$); point lands in trivial subgroup. $\mathbb{Z}/2$: only point hits 0 .
Ab, Vect _k , R-Mod, Mon	✗	the zero/identity element	zero object $\Rightarrow$ point blind to all but 0 .
Set* (pointed sets)	✗	the basepoint	zero object = basepoint.
Ring/CRing (unital)	✗	hom from zero ring $\Rightarrow$ forces $1=0$	nontrivial rings have no points at all .
G-Set	✗	a fixed point ($g \cdot a = a \ \forall g$)	A^G rarely covers; free action has none. A topos, yet not WP.
Presheaves $\text{Set}^{C^{op}}$	✗	a global section	global sections miss local data.
Sheaves $\text{Sh}(X)$	✗	a global section	non-constant sheaf's stalks unreachable.

Three ways to fail, one to succeed:

1. **Zero object** ($1 \cong 0$) $\rightarrow$ only point is identity/zero (Grp, Ab, Vect, Mod, Mon, Set*).
2. **Terminal forces triviality** $\rightarrow$ nontrivial objects have *no* points (Ring).
3. **Points = invariant/global data** missing local structure (G-Set fixed points, sheaf/presheaf global sections).

Succeeds when points *are* the elements and elements separate everything: **Set** + well-pointed topoi.

§5. Well-supported vs well-pointed: capital (§1.525)

§3–§4 are about **well-pointed** objects. The opposite gap — objects that are **well-supported** yet **not** well-pointed — is what capitalization repairs, and naming the categories without that gap is §1.525.

Definition 5.1 (Capital (§1.525)).

A pre-regular category is **capital** if every well-supported object is well-pointed. Equivalently (§1.526), the terminator **1** is projective — and then $\Gamma = \text{Hom}(1, -)$ is a representation of pre-regular categories.

Is “**capital**” standard terminology? No. It is Freyd & Scedrov’s own coinage, like *logos*, *allegory*, *tabular*, *terminator*; it is not used with this meaning elsewhere. The standard names for the same property are “1 is (regular-)projective”, “1 is a projective generator”, or “ $\Gamma = \text{Hom}(1, -)$ preserves covers”. Read the word as **rich in points**: Γ is the points / global-sections functor, and a category is **capital** when every supported object is reached by its points. (Freyd writes the capitalization with an underline, $\underline{A}$; the name attaches to the process of enriching $\underline{A}$ with points, not to capital letters.)

§5.1. Two $\mathbb{Z}/2$ ’s — and each as a category

The $\mathbb{Z}/2$ that is *not projective* in **Ab** (a **group**) and the $\mathbb{Z}/2$ -set that is *not well-pointed* in **G-Set** (a **set with an action**) share the carrier $\{0, 1\}$, but they are different objects — and, viewed as categories, different categories. Write the two elements of $\mathbb{Z}/2$ as **id** and σ (σ the generator, $\sigma\sigma = \text{id}$; additively $\text{id} = 0$, $\sigma = 1$).

A $\mathbb{Z}/2$ -set is not *itself* a category — it is a set with an action, equivalently a functor from the one-object category of $\mathbb{Z}/2$ to **Set**. To turn either $\mathbb{Z}/2$ into a category:

view	objects	morphisms
the group $\mathbb{Z}/2$ as a one-object category	one, $*$	$\{\text{id}, \sigma\}$; σ is a self-iso $* \rightarrow *$, $\sigma\sigma = \text{id}$
the swap set as action groupoid $\{0, 1\} // \mathbb{Z}/2$	two, 0 and 1	id_0, id_1 , and an iso $\sigma : 0 \rightarrow 1$, $\sigma : 1 \rightarrow 0$ (σ swaps), $\sigma\sigma = \text{id}$

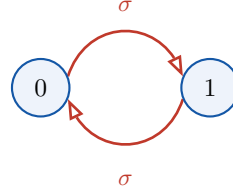
In the action groupoid a morphism $a \rightarrow b$ is a group element g with $g \cdot a = b$; since σ swaps $0 \leftrightarrow 1$ it gives the iso $0 \cong 1$, while each object’s only endomorphism is its identity. (The swap is translation by σ — $\mathbb{Z}/2$ acting on its *own* underlying set, the regular action — which is why this groupoid is a torsor.)

the GROUP $\mathbb{Z}/2$ (one object)

id


 one object, a self-iso ($\sigma\sigma = \text{id}$)

not contractible

 the swap $\mathbb{Z}/2\text{-SET}$ (action groupoid)

 two objects, an iso $0 \cong 1$

 contractible ($\cong$ a point); a torsor

Figure 6: Each $\mathbb{Z}/2$ as a category. **Left:** the group $\mathbb{Z}/2$ as a one-object category — **one** object $*$ with a **self-iso** σ ($\sigma\sigma = \text{id}$); a self-loop encodes a group element, so it is not contractible. **Right:** the swap $\mathbb{Z}/2$ -set as its action groupoid $\{0, 1\}/\mathbb{Z}/2$ — **two** objects $0, 1$ with an iso σ **between** them; an iso between distinct objects only makes them isomorphic, so it collapses to a point. (This depicts the category, not a commutative diagram: $\sigma \neq \text{id}$, though both run $* \rightarrow *$.)

One object or two? Two. “One object with a self-iso” is the *group* as a one-object category, a different category: a self-loop $\sigma : * \rightarrow *$ encodes a group element, so the object’s automorphism group is $\mathbb{Z}/2$, and it is not contractible (its single object has a nontrivial automorphism). An iso $0 \rightarrow 1$ between *distinct* objects only says they are isomorphic, and that groupoid collapses to a point.

Used in the next subsections. A fixed point of the action is an object with a nontrivial self-iso ($g \cdot a = a$). The swap has none, so no object of its action groupoid carries a self-iso — exactly why it is **two** objects, not one. (And in **Ab** the group $\mathbb{Z}/2$ has its single point trapped at $\text{id} = 0$; in **G-Set** the swap set has no point at all.)

§5.2. A well-supported object that is not well-pointed

Recall **well-supported** = $A \rightarrow 1$ is a cover, and **well-pointed** = the points $1 \rightarrow A$ jointly cover A . A **non-capital** category has an object with the first but not the second. Three witnesses:

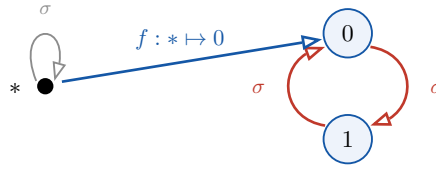
Finite witness — a $\mathbb{Z}/2$ -set with no fixed point. In **G-Set** with $G = \mathbb{Z}/2 = \{e, \sigma\}$, take $A = \{0, 1\}$ with σ acting by **swap** ($\sigma \cdot 0 = 1, \sigma \cdot 1 = 0$); the terminal $1 = \{*\}$ carries the trivial action.

- **Well-supported:** $A \rightarrow 1$ is surjective, hence a cover. ✓
- **Not well-pointed:** a point $1 \rightarrow A$ is equivariant, so its value must be a **fixed point** of A ; the swap has none, so $\text{Hom}(1, A) = \emptyset$. With no points, the smallest subobject containing every point is $\emptyset \subsetneq A$. ✗

So A surjects onto 1 , yet its points see nothing of it. **G-Set** is a topos, **yet not capital**.

Why can’t you just pick 0 or 1 ? You *can* pick an element of the underlying set — but the result is a bare function $* \mapsto 0$, not a **G-set** morphism. A point $1 \rightarrow A$ must be **equivariant** ($f(g \cdot x) = g \cdot f(x)$), and the terminal $1 = \{*\}$ carries the **trivial** action ($\sigma \cdot * = *$). So equivariance forces $f(*) = \sigma \cdot f(*)$ — the chosen element must be **fixed**. Take $f(*) = 0$: then $f(\sigma \cdot *) = f(*) = 0$, but $\sigma \cdot f(*) = \sigma \cdot 0 = 1$, so it would need $0 = 1$. Picking 1 fails the same way; the swap fixes nothing, so no choice is equivariant and $\text{Hom}(1, A) = \emptyset$ — exactly the **Ab** pattern, where a bare element-choice fails to be a homomorphism.

object $\mathbf{1}$ (one point $*$), σ trivial object A : points $0, 1$, σ swaps



equivariance needs $f(\sigma \cdot *) = \sigma \cdot f(*)$, i.e. $0 = 1$ ✗

Figure 7: Why no element is a point of the swap set. The candidate $f : * \mapsto 0$ is a function but not a $\mathbf{G}$ -map: $\mathbf{1} = \{*\}$ has the trivial action ($\sigma \cdot * = *$), so $f(\sigma \cdot *) = f(*) = 0$, whereas $\sigma \cdot f(*) = \sigma \cdot 0 = 1$; equivariance would force $0 = 1$. Picking 1 fails identically — the swap fixes nothing, so $\text{Hom}(\mathbf{1}, A) = \emptyset$.

Two more, one line each:

- **Ab / Grp**: any nonzero object, e.g. $\mathbb{Z}/2$. It surjects onto the terminal (the zero group), so it is **well-supported**; but the only point $\mathbf{1} \rightarrow \mathbb{Z}/2$ is 0 , and $\{0\} \subsetneq \mathbb{Z}/2$ already holds every point — **not** well-pointed. (Here $\mathbf{1} \cong 0$: the zero object traps the point.)
- **Sheaves**: a nontrivial $\mathbb{Z}/2$ -torsor on the circle (the connected double cover) is locally inhabited, so $F \rightarrow \mathbf{1}$ is a cover (**well-supported**), but it has **no global section at all** — **not** well-pointed. The geometric face of the same phenomenon.

★ Punchline

Capitalization adjoins the missing points: for each well-supported A that is not well-pointed it adds a point $\mathbf{1} \rightarrow A$ in a larger A^* (the slice over A , §1.546), repeating until **every** well-supported object is well-pointed — i.e. until the category is **capital**. That is the §1.54 machine the rest of these notes build toward.

§5.3. Is the terminator $\mathbf{1}$ always projective? No

A natural guess: $\mathbf{1}$ is always projective, since a cover $A \twoheadrightarrow \mathbf{1}$ ought to split. It does in **Ab** — your $\mathbb{Z}/2 \rightarrow \mathbf{1}$ splits by the zero map $0 \rightarrow \mathbb{Z}/2$ — but that is special, not general.

Definition 5.3.1 (When $\mathbf{1}$ is projective).

$\mathbf{1}$ is **projective** $\iff$ every cover $A \twoheadrightarrow \mathbf{1}$ has a section $\mathbf{1} \rightarrow A \iff$ **every well-supported object has a point**. Capital (§1.525) supplies this (`capital_implies_one_projective` in `S1_52.lean`). It is **not** automatic.

Category	$\mathbf{1}$ projective?	Why
Ab , Grp	✓	zero object: the canonical point $0 \rightarrow A$ (your $0 \rightarrow \mathbb{Z}/2$) splits $A \rightarrow \mathbf{1}$.
SET	✓	$A \rightarrow \mathbf{1}$ a cover $\implies A$ nonempty $\implies$ a point to choose (AC).
Ring , CRing	✗	terminal $\mathbf{1} =$ zero ring (no zero <i>object</i>); a nontrivial R is well-supported but $\text{Hom}(\mathbf{1}, R) = \emptyset$ — a hom out of the zero ring forces $\mathbf{1} = 0$.
G-Set	✗	the swap $\mathbb{Z}/2$ -set is well-supported but has no fixed point $\implies$ no point $\mathbf{1} \rightarrow A$.

Category	1 projective?	Why
Sheaves	✗	a torsor with no global section: well-supported, but $\Gamma = \emptyset$.

★ Punchline

If **1** were *always* projective, capitalization would do nothing. Its whole job (§1.54) is to enlarge a non-capital category until every well-supported object acquires a point — i.e. until **1** becomes projective. Your **Ab** example works only because the terminal there is a *zero* object.

§6. “Set-like, and structure breaks it” — sharpened

The instinct (WP = Set-like; add structure, lose it) is right, with one correction: it’s not *any* structure — it’s structure that **constrains maps out of 1** so points stop reaching every element. In Set, **1** is a *bare* point (maps anywhere). Structure replaces **1** with a *specific* one-element object whose maps out are constrained:

- a **marked element** (identity/zero/basepoint) traps the point;
- **operations/morphisms tying elements together** (action, topology, restriction maps, an arrow) $\Rightarrow$ points = the globally-coherent/invariant part, missing local structure.

But **parallel** structure (no binding operation) stays WP:

Set $\times$ Set: point of (A,B) is (a,b) $\in A \times B$ $\rightarrow$ covers. $\checkmark$ WP
 Set $^{\rightarrow}$ (one arrow $f:X \rightarrow Y$): point is $x \in X$ ($f(x)$ forced) $\rightarrow$ misses $Y \setminus \text{im}(f)$. $\times$ not WP

Same two sets — adding *a morphism between them* breaks it. Breaker = **binding** structure.

Clean statement. WP $\iff$ **1** is a bare point hitting every element (**1** is a generator/separator by covers). Programmer view: an object is WP iff `obj.points()` exposes everything; binding structure adds **hidden state** (a forced identity, a fixed-point requirement, local data) points can’t see.

§7. Is it useful / does modern math have it?

It’s a **characterization**, not a property you hope for — it pins down *when point-wise reasoning is valid*. Its narrowness (as stated, **1 alone** generates) is informative: it’s the boundary of “Set-like.” Weaken to “**some** family generates” and it’s ubiquitous:

Modern name	Where	Meaning
Well-pointed topos	Lawvere ETCS , categorical set theory	1 generator + 2-valued + nontrivial $\Rightarrow$ a model of set theory
Generator / separator	module/abelian/Grothendieck categories	$\text{Hom}(G, -)$ faithful; Grothendieck cats <i>defined</i> by having one

Modern name	Where	Meaning
Enough points	topos/locale theory	points detect everything; some topoi have none
Enough global sections	algebraic geometry	ampleness/affineness — $\Gamma = \text{Hom}(1, -)$ sees the sheaf
Concretizable	general cat theory	faithful functor to Set; Freyd: $\text{Ho}(\text{Top})$ is not

§8. “Just embed into Set” — three problems

1. **Sometimes impossible:** Freyd, “*Homotopy is not concrete*” — $\text{Ho}(\text{Top})$ has no faithful functor to Set.
2. **Faithful isn’t enough:** transporting a proof needs the functor to **preserve** the structure chased (finite limits, monos, images, exactness = **exact/regular**) *and* **reflect** it (**conservative**). Bare faithfulness gives neither.
3. **Building such a functor is the theorem:** Freyd–Mitchell (abelian $\rightarrow \text{R-Mod} \rightarrow \text{Set}$), Barr (regular $\rightarrow \text{Set}$). Capitalization is the machine that builds it. Not free.

Concrete vs abstract. Grp comes with a forgetful $\text{U: Grp} \rightarrow \text{Set}$ that is faithful, limit-preserving, conservative — so element-chasing in groups is free. The problem is **abstract** categories (abstract abelian/regular/topos) with **no** underlying-set functor; you must construct one.

Yoneda alternative. $\mathbf{A} \hookrightarrow [\mathbf{A}^{\text{op}}, \text{Set}]$ always, fully faithfully — but into **presheaves** (not WP), giving **generalized** elements (maps from representables, the functor-of-points), not honest points.

§9. Why Yoneda isn’t enough

Two reasons, one shallow, one deep:

- **Shallow:** target is presheaves, not well-pointed $\Rightarrow$ generalized elements, not points from 1.
- **Deep:** Yoneda is **continuous** (preserves & reflects limits) but **destroys colimits/exactness**. In $\mathbf{Ab}$: $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ is epi, but $\text{Hom}(-, \mathbb{Z}) \rightarrow \text{Hom}(-, \mathbb{Z}/2)$ is **not** epi in presheaves (a map $X \rightarrow \mathbb{Z}/2$ needn’t lift to $X \rightarrow \mathbb{Z}$). So an exact sequence need not stay exact $\Rightarrow$ a chase in presheaves proves nothing about $\mathbf{A}$. Yoneda is only *half*-exact.

	full+faithful	limits	covers/colimits	into well-pointed Set
Yoneda (free)	✓	✓	✗	✗ (presheaves)
Freyd–Mitchell / Barr (theorem)	✓	✓	✓	✓

The two checkmarks Yoneda lacks — preserving covers, reaching honest points — are exactly the embedding theorems’ content.

§10. Why capitalization preserves exactness when Yoneda doesn’t

The difference is **how points are added**.

- **Yoneda probes by mapping into objects** ($a \mapsto \text{Hom}(-, a)$). $\text{Hom}(X, -)$ is intrinsically limit-preserving / colimit-destroying $\Rightarrow$ left-exact only; and it **leaves** the regular world for presheaves.
- **Capitalization stays inside the regular world and adds points by slicing**: each rung is the pullback functor $\Delta : A \rightarrow A/B$, $X \mapsto (X \times B \rightarrow B)$ (base change along $B \rightarrow 1$).

The decisive fact is the **defining axiom of a regular category** (§1.52): “**pullbacks transfer covers**” — covers are stable under pullback. So the slice functor, *being a pullback*, **automatically preserves covers**, and (being a pullback) preserves finite limits. Finite-limits + covers = a **representation of regular categories** $\Rightarrow$ preserves images and **exactness**. Faithful when B is well-supported (§1.532).

Capitalization is a transfinite tower $A \rightarrow A/B \rightarrow (A/B)/B' \rightarrow \dots$ of such rungs, each exact + faithful, with colimit $A \subset A^*$ an **exact faithful** embedding that makes well-supported objects well-pointed. Exactness is **inherited from pullback-stability of covers** — the very axiom that makes A regular.

★ Punchline

Yoneda probes with Hom (left-exact, leaves the world); capitalization enlarges with **pullback** (cover-stable by the regularity axiom, fully exact, stays in the world). “Covers pull back” is exactly what Hom -based Yoneda never uses and what makes slice-based point-adding exact.

(Repo: `Slice.lean` Σ reflects covers; `SliceRegular.lean` slice is pre-regular; `RelativeCapitalization.lean` the per- B rung $A \rightarrow A/B$ as a faithful, point-acquiring `CapStep`.)

§11. The construction concretely: Δ , rungs, “different B ”, faithful-not-full

How capitalization actually adds the points (§1.525–§1.547), with the Set pictures.

§11.1. Two stacked indexings — “rung” vs “layer”

- **Rung** = one slice step $A \xrightarrow{\Delta} A/B$. Inside one *relative capitalization* (§1.546) you take a transfinite ascending union of rungs $A = A_0 \subset A_1 \subset \dots$, where each rung slices over **the first well-supported object B not yet pointed** (first w.r.t. a fixed well-order on objects). **B changes every rung**. When the layer closes, every well-supported object of the *original* A has its points.
- **Layer** = one relative capitalization. The outer tower $A \subset A^* \subset A^{**} \subset \dots$ (§1.545) stacks layers, because A^* has **new** well-supported objects (built from the new points) that themselves need points. The union A_ω of the ω -tower is **capital** (every well-supported object is well-pointed).

You do **not** iterate $A/B \rightarrow (A/B)/B$ on one fixed B ; and nothing shrinks — every step is a faithful *enlargement*, the towers are ascending unions climbing up to the richest category A_ω .

§11.2. What Δ is (§1.546 names the functor $A \rightarrow^{A/B} A/B$)

$\Delta : A \rightarrow A/B$, $X \mapsto (X \times B \xrightarrow{\text{snd}} B) = \text{base change along } B \rightarrow 1$: since $A = A/1$, pulling the object X (i.e. $X \rightarrow 1$) back along the unique $B \rightarrow 1$ gives $X \times B \rightarrow B$. Repo: `RelativeCapitalization.lean` `sliceEmbedObj B C := (prod C B, snd)` is exactly ΔC (named `sliceEmbed*`, not Δ).

In HASK (the `Set` model, B a fixed type) Δ is two one-liners — a slice object over b is a carrier with a structure map $c \rightarrow b$, and Δ produces the carrier (x, b) with `snd`:

```
-- Δ : Set → Set/B    (base change along B → 1; the slice embedding A → A/B)
deltaObj :: (x, b) -> b    -- on objects:  X ↦ (X×B, snd);
carrier is (x, b)
deltaObj = snd
deltaMor :: (x -> y) -> (x, b) -> (y, b)    -- on morphisms: f ↦ f × id_B, a map
over B
deltaMor f (x, b) = (f x, b)
```

It is a functor (`deltaMor id = id`, `deltaMor (g . f) = deltaMor g . deltaMor f`), **faithful** iff b is inhabited (B well-supported), and the generic point of ΔB is the diagonal $\backslash b \rightarrow (b, b)$.

Why it adds a point: in A/B the terminal is $(B \xrightarrow{\text{id}} B)$; a point of $\Delta B = (B \times B \xrightarrow{\text{snd}} B)$ is a section of `snd`, and the **diagonal** $\delta = \langle \text{id}, \text{id} \rangle$ is one ($\delta \gg \text{snd} = \text{id}$). Through `fst`, δ is the identity element of B — a “generic point of B ” that didn’t exist in A . Base-change functors preserve finite limits and (regular axiom) covers $\Rightarrow \Delta$ is an exact representation (§1.543).

§11.3. Two A/B ’s: the slice (§1.26) vs the redefinition (§1.544)

$X \times B$ is not in the definition of A/B . §1.26 defines A/B as **all arrows into B** — in `Set` (§1.261) the general object is a B -indexed family of fibers $\{A_i\}$, fibers arbitrary. The bundle $\Delta X = X \times B \rightarrow B$ is just the **constant** family (every fiber $= X$), a thin sub-population of A/B produced by the **functor** Δ — nothing the slice’s definition mentions.

For capitalization the book **redefines** A/B so that A is a literal subcategory (§1.544), via the **inflation** A' — products strictified into lists (the free-monoid trick):

- objects of $A' =$ finite **lists** $\langle A_1, \dots, A_n \rangle$ of A -objects (“formal products”); the forgetful $A' \rightarrow A$ sends a list to a chosen product $A_1 \times \dots \times A_n$ ($\langle \rangle \mapsto 1$). Hom-sets are A' ’s, so $A' \simeq A$.
- product in $A' =$ **list concatenation** — strictly associative, with **strict cancellation** $B \times A = B \times A' \Rightarrow A = A'$ (drop the prefix).

Then $\Delta(X) = \langle X, B \rangle$ (append B), and the book finally **relabels** Δ ’s image $\langle X, B \rangle$ as X itself, so $A \subseteq A/B$ on the nose. The lists force injectivity: even when B is not well-supported and $B \times U \cong B \times 1$ **collapses in A** (§1.534), the lists $\langle U, B \rangle \neq \langle 1, B \rangle$ stay distinct — so Δ **separates objects** always, and **separates morphisms** (is faithful) only when B is well-supported.

Pictorially, “inflation” is the general §1.36 move, an instance of the §1.243 **founding** construction — take a **class** of labels with an **underlying function** to the base, and **borrow the morphisms** from the base (formalized as `FoundingData` / `inflation` in `S1_35.lean`):

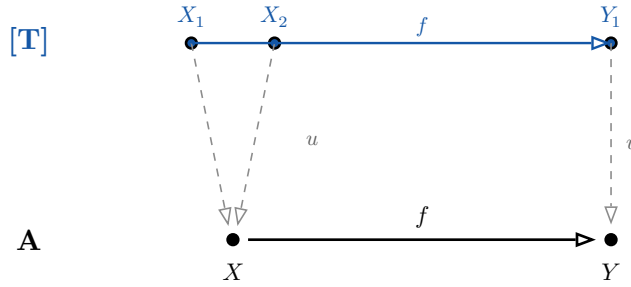


Figure 8: The §1.36 **inflation** $[T]$ of A — the §1.243 founding with the **most inclusive** predicate ($\text{Hom}' =$ all proto-morphisms). Objects are **labels** (the class $\mathcal{C} = \{X_1, X_2, Y_1\}$); the **underlying function** $u : \mathcal{C} \rightarrow |A|$ (dashed) sends each label to its base object — $u X_1 = u X_2 = X$, $u Y_1 = Y$. A morphism is **borrowed from** A : $\text{Hom}_{[T]}(X_1, Y_1) = \text{Hom}_A(X, Y) = \{f\}$, so the **one** arrow f serves both $X_1 \rightarrow Y_1$ and $X_2 \rightarrow Y_1$. Forgetting the labels (u) gives back A , so $[T] \simeq A$. (In §1.544 the labels are lists $\langle A_1, \dots, A_n \rangle$ and u multiplies them out to products.)

§11.4. Δ is faithful, not full — and that is “slicing adds points”

Set picture, $B = \{0, 1\}$ so $\text{Set}/B \cong \text{Set} \times \text{Set}$, $\Delta : X \mapsto (X, X)$, $f \mapsto (f, f)$:

- **Faithful** (loses no morphism): $f \mapsto (f, f)$ is injective — provided B is well-supported ($\text{Hom}(X, Y) \hookrightarrow \text{Hom}(X \times B, Y)$, $f \mapsto f \circ \text{fst}$, injective iff fst epi iff $B \rightarrow 1$ a cover; that’s *why* you only slice over well-supported B , never over $\emptyset$).
- **Not full** (the slice is strictly bigger): $\text{Hom}_{\text{Set}/B}(\Delta X, \Delta Y) = \text{Hom}(X, Y)^{\wedge \{B\}}$, and Δ hits only the **diagonal** $\text{Hom}(X, Y) \hookrightarrow \text{Hom}(X, Y)^{\wedge \{B\}}$. Full $\iff B \cong 1$. Capitalization slices over $B \not\cong 1$, so Δ is *never* full there.

Minimal witness $\Delta 1 \rightarrow \Delta Y$, $Y = \{0, 1\}$ — a map over B = a section of ΔY = one Y -value per fiber, **independently**:

$$\begin{array}{ccc} \Delta Y = (Y \times B \rightarrow B), & \text{fibers } Y \mid Y & \text{maps } \Delta 1 \rightarrow \Delta Y : \begin{array}{c} (0, 0) \quad (1, 1) \mid (0, 1) \quad (1, 0) \\ Y^2 = 4 \end{array} = \\ & \text{over } 0 \quad \text{over } 1 & \begin{array}{c} \text{ } \Delta \text{'s image } \downarrow \text{ } \Delta \text{ NOT in image } \downarrow \\ \text{"same both fibers"} \quad \text{"differs per fiber"} \end{array} \end{array}$$

The 2 off-diagonal maps are **not** Δh for any point $h : 1 \rightarrow Y$ — *and* they are exactly the **new points** of ΔY (Y had 2 points in Set ; ΔY has $2^2=4$ in Set/B). So “ Δ not full” = “slicing added points”. **No zero object is involved** — Set has none ($\emptyset \neq \{*\}$); the extra freedom is per-fiber independence, not anything mapping to $\emptyset$.

§11.5. Different B ’s “jointly prove something” (§1.547, §1.526)

One B gives B points; ranging over **all** B gives every object points; the union assembles the capital A_- , where $\Gamma = \text{Hom}(1, -) : A_- \rightarrow \text{Set}$ is **exact** (capital $\implies 1$ projective, §1.526). Faithful is *joint*: the family $T_B : A \rightarrow^{\Delta} A/B \rightarrow A_-/B \rightarrow^{\Gamma} \text{Set}$ (one per B) together separates morphisms and detects every proper subobject — no single B suffices. That faithful exact $A \hookrightarrow A_- \rightarrow \text{Set}$ is the representation Yoneda couldn’t give. The outer embedding $A \hookrightarrow A_-$ is itself only **faithful**, not full (book claims faithful, §1.544) — same faithful-not-full pattern as each rung.